Dually Supervised Track-Before-Detect Processing of Multichannel Video SAR Data

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***Abstract*— Track-before-detect (TBD) algorithm has been used to track weak moving target shadows in video synthetic aperture radar (SAR), where strong maneuverability may deteriorate tracking performance. Fortunately, the Doppler characteristic of the target can additionally provide velocity guidance for tracking. This article presents ground moving target indication (GMTI) results of the multichannel video SAR raw data, and a tracking approach is proposed based on the Doppler supervision in an improved dynamic programming-based TBD (DP-TBD) framework that uses a dual-domain merit function. Both the sequential high-resolution SAR images and low-resolution range-Doppler (RD) spectra are used, and the clutter in the RD spectrum is suppressed by using an adaptive displaced phase center antenna (ADPCA) technique. Fine states can be searched in dual-domain through resampling from random expansion in state initialization and transition. The inverse shadow amplitude and Doppler energy of the same potential target are simultaneously integrated to determine whether the target exists. The proposed Dual-DP-TBD can deal with the tracking of a time-varying number of targets in successive measurements. Compared to other TBD algorithms, this approach has fewer missing alarms and false alarms, thanks to precise velocity estimation constrained from diverse features both in SAR image and RD spectrum.**

***Index Terms*— Ground moving target indication (GMTI), radar imaging, synthetic aperture radar (SAR), track-before-detect (TBD), video SAR.**

I. Introduction

VIDEO synthetic aperture radar (SAR) can provide dynamic surveillance for the area of interest and slow ground moving targets. It has shown great potential in moving target tracking. A few video SAR systems have been developed recently [1]–[5]. In particular, imaging and tracking using an unmanned aerial vehicle (UAV)-borne W-band radar have been reported in [5]. Some efforts in video SAR image formation have been published [6]–[9].

Video SAR is developed essentially for ground moving target detection and tracking. Displaced phase center antenna (DPCA) and along-track interferometry (ATI) techniques have been widely used in multichannel radar systems to suppress strong clutter [10]–[12]. However, the target energy is always smeared and shifted in SAR image, and as a result, it is difficult to directly detect and relocate the moving target due to the limited signal-to-clutter-and-noise ratio (SCNR). Meanwhile, the channel mismatch is inevitable in practice, which requires extra calibration and equalization [13]–[15].

It has been found that a moving target may leave a shadow at its real position due to obstruction on incident radar radiation and Doppler shift of its energy [16], particularly in the case of short beam dwell time of video SAR. The current shadow-based detection methods mainly include conventional implementations [17]–[19] and deep learning-based approaches [20]–[22]. However, shadow detection becomes difficult in some deteriorating cases, such as small-size targets. In addition, the shadow expands along the target moving direction, resulting in blurred boundaries and poor contrast, especially for the fast-moving target. Therefore, moving target shadow in video SAR is generally a typically weak and small target.

The track-before-detect (TBD) algorithm has been widely used in dim target detection and tracking under complex clutter environments [23]. The dynamic programming-based TBD (DP-TBD) is a representative TBD technique [24]–[26]. In long-time tracking, the multiframe (MF) DP-TBD algorithm usually uses a sliding window (SW) strategy. However, traditional SW-DP-TBD brings repeated calculations due to overlapping frames in two adjacent windows. Therefore, an efficient recursive MF-TBD (R-MF-TBD) algorithm was proposed [25], which adopts a recursive merit function to avoid redundant calculation. The TBD algorithm for shadow detection in video SAR using the expanding and shrinking strategies has been presented in [27].

However, there still exist many difficulties in shadow-based detection by TBD, particularly in the tracking of maneuvering targets. First, it is difficult to obtain the accurate position of shadows in the preliminary detection [27], resulting in heavy errors in the initial state, especially for the velocity components. Second, for the conventional state transition using quantified acceleration process noises, the small number of valid transition states is not sufficient to track the maneuvering target while a large number of states increase the computational load and the expanded states may cover other areas of weak reflectivity in the image, resulting in poor tracking performance. It can be concluded that neither direct energy detection nor shadow-based detection can be individually relied on for robust detection in video SAR. Therefore, the joint detection scheme by combining both energy and shadow has been presented in our previous works [28], [29].

This article presents a new dually supervised TBD approach by embedding the Doppler supervision into an improved DP-TBD algorithm, termed Dual-DP-TBD, and shows the

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processing results of UAV-borne multichannel measured video SAR data. Both the sequential high-resolution video SAR images and low-resolution range-Doppler (RD) spectra are used to improve ground moving target indication (GMTI) performance. Compared to other classical TBD algorithms, the proposed Dual-DP-TBD approach has fewer missing alarms and false alarms. The main contributions of this article include the following.

1) An adaptive DPCA (ADPCA) approach is presented to improve the SCNR in the preprocessing. The stationary clutter and noise are suppressed by using preequalization strategy and masked adaptive optimization.
2) We propose a fine state estimation method by using Doppler supervision, which provides effective velocity guidance. Fine initial or transferred states can be searched in dual-domain through resampling from random expansion in state initialization and transition. This new state transition method does not depend on the conventional quantified acceleration process noises.
3) The designed dual-domain merit function is used to simultaneously integrate inverse shadow amplitude in SAR image and Doppler energy in RD spectrum. Diverse features are joined to determine whether the target exists. Additionally, the proposed approach can realize the detection and tracking of a time-varying number of targets, such as known, newborn, and early-exiting targets.

This article is organized as follows. Section II discusses the ADPCA approach. Section III details the proposed Dual-DP-TBD approach. Section IV shows the processing results of measured video SAR data. Section V concludes this article.

II. Multichannel Clutter Suppression

A. Classical Channel Equalization and DPCA

DPCA has been used to suppress clutter in the GMTI system of airborne radar. However, the channel mismatch is an important concern, which may deteriorate the performance a lot. The channel mismatch can be modeled as the errors along the fast-time and slow-time frequencies, which is due to the changes in temperature and supply voltage that affect the fast-time sampling and platform motion error and pattern error that destroy the amplitude and phase consistencies between the channels over slow time, respectively. The channel characteristic can be balanced by solving the following optimization problem [15]:

$$\min_{\boldsymbol{H}_{\mathrm{rc}},\boldsymbol{D}_{\mathrm{rc}}} \int\int \left|\boldsymbol{S_r} - \boldsymbol{H}_{\mathrm{rc}} \cdot \boldsymbol{S_c} \cdot \boldsymbol{D}_{\mathrm{rc}}\right|^2 df_r df_a \tag{1}$$

where $\boldsymbol{H}_{\mathrm{rc}}$ and $\boldsymbol{D}_{\mathrm{rc}}$ are compensation items along the fast-time frequency f_r and slow-time frequency f_a, and $\boldsymbol{S_r}$ and $\boldsymbol{S_c}$ are aligned return matrices in 2-D frequency domain obtained from the reference and calibration channels, respectively. Symbol $\cdot$ represents the Hadamard product. Theoretically, the aligned two-channel returns have the same spatial position in each slow-time index but with a time interval $\Delta t = B_x/v_a$, where B_x is the baseline length and v_a is the azimuth velocity of platform, as shown in Fig. 1.

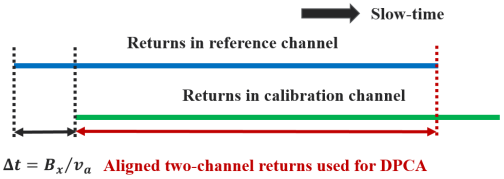


Fig. 1. Aligned two-channel returns used for DPCA.

Moreover, (1) can be solved by the following iteration [15]:

$$\begin{aligned}
\tilde{\boldsymbol{S}}_{\boldsymbol{c}(j+1)} &= \tilde{\boldsymbol{S}}_{\boldsymbol{c}j} \cdot \boldsymbol{T}_1, \quad \boldsymbol{T}_1 = \frac{\int \tilde{\boldsymbol{S}}^*_{\boldsymbol{c}j} \cdot \boldsymbol{S_r} df_r}{\int \left|\tilde{\boldsymbol{S}}_{\boldsymbol{c}j}\right|^2 df_r} \\
\tilde{\boldsymbol{S}}_{\boldsymbol{c}(j+2)} &= \tilde{\boldsymbol{S}}_{\boldsymbol{c}(j+1)} \cdot \boldsymbol{T}_2, \quad \boldsymbol{T}_2 = \frac{\int \tilde{\boldsymbol{S}}^*_{\boldsymbol{c}(j+1)} \cdot \boldsymbol{S_r} df_a}{\int \left|\tilde{\boldsymbol{S}}_{\boldsymbol{c}(j+1)}\right|^2 df_a}
\end{aligned} \tag{2}$$

where * is conjugate operation, $\tilde{\boldsymbol{S}}_{\boldsymbol{c}j}$ is calibrated return matrix in the jth iteration. $\tilde{\boldsymbol{S}}_{\boldsymbol{c}0} = \boldsymbol{S_c}$, and $j = 0, 2, 4, \ldots$. The iteration stops until ratio terms $\boldsymbol{T}_1$ and $\boldsymbol{T}_2$ approximately equal to $\mathbf{1}$.

After channel equalization (CE), the amplitude and phase consistencies between two-channel returns are improved. Since the RD spectrum can provide velocity guidance for target tracking, the DPCA method is used in the RD domain and the suppressed the RD spectrum $\boldsymbol{R_d}$ can be expressed as

$$\boldsymbol{R_d} = \boldsymbol{R_{dr}} - \boldsymbol{R_{dc}} \tag{3}$$

where $\boldsymbol{R_{dr}}$ and $\boldsymbol{R_{dc}}$ are reference and calibrated RD spectra, respectively. Theoretically, stationary clutter can be eliminated while the moving target can be distinguished.

B. Adaptive DPCA

The differences between two channels over all range and azimuth frequencies are compensated by (2). However, some frequencies contain the moving target whose inherent differences caused by time interval Δt should not be calibrated. Unfortunately, these differences are destroyed due to the entire integration in (2). Hence, the moving target may be eliminated during DPCA processing and its SCNR even deteriorates. An ADPCA approach is proposed for the 2-D dechirp signal to effectively suppress stationary clutter and improve the target's SCNR, which is very useful for the resampling of fine state and Doppler energy integration in the subsequent Dual-DP-TBD algorithm, detailed as follows.

1) Preequalization for Strong Stationary Clutter: For 2-D dechirp returns, fast-time and slow-time frequencies correspond to different ranges and Doppler, respectively, and thus scatterers' returns can be preliminarily focused in the 2-D frequency domain. To suppress the ground clutter, a preequalization strategy is used, which can be expressed as

$$\begin{aligned}
\bar{\boldsymbol{S}}_{\boldsymbol{c}(j+1)} &= \bar{\boldsymbol{S}}_{\boldsymbol{c}j} \cdot \frac{\sum_{i=1}^{F_N} \bar{\boldsymbol{S}}_{\boldsymbol{c}j}(f_{ri}, f_a)^* \cdot \boldsymbol{S_r}(f_{ri}, f_a)}{\sum_{i=1}^{F_N} \left|\bar{\boldsymbol{S}}_{\boldsymbol{c}j}(f_{ri}, f_a)\right|^2} \\
\bar{\boldsymbol{S}}_{\boldsymbol{c}(j+2)} &= \bar{\boldsymbol{S}}_{\boldsymbol{c}(j+1)} \cdot \frac{\sum_{i=1}^{F_M} \bar{\boldsymbol{S}}_{\boldsymbol{c}(j+1)}(f_r, f_{ai})^* \cdot \boldsymbol{S_r}(f_r, f_{ai})}{\sum_{i=1}^{F_M} \left|\bar{\boldsymbol{S}}_{\boldsymbol{c}(j+1)}(f_r, f_{ai})\right|^2}
\end{aligned} \tag{4}$$

where $\bar{\boldsymbol{S}}_{\boldsymbol{c}j} \in \mathbb{C}^{N_r \times N_d}$ is the precalibrated return matrix in the jth iteration. $\bar{\boldsymbol{S}}_{\boldsymbol{c}0} = \boldsymbol{S_c}$ and $j = 0, 2, 4, \ldots$ N_r and N_d are

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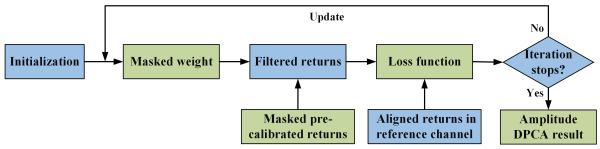


Fig. 2. Flowchart of DPCA with adaptive optimization.

the number of range bins and Doppler channels, respectively. F_N is the number of fast-time frequencies that need to be calibrated (likewise for F_M). All F_M slow-time frequencies that belong to clutter band should be calibrated. All resolution units in $\boldsymbol{S_c}$ are sorted in descending order according to their amplitudes and the fast-time frequencies of top F_N strong points are selected. Therefore, F_N fast-time frequencies and F_M slow-time frequencies mainly contain the returns of strong stationary clutter.

2) Mask for Potential Moving Target: After preequalization, the main stationary clutter is suppressed in $\bar{\boldsymbol{R}}_{\boldsymbol{d}} \in \mathbb{C}^{N_r \times N_d}$, where $\bar{\boldsymbol{R}}_{\boldsymbol{d}} = \boldsymbol{S_r} - \bar{\boldsymbol{S}}_{\boldsymbol{c}}$. To improve the target's SCNR, the residual clutter and noise should be further suppressed. Here, the mask strategy is adopted to avoid the elimination of the moving target in subsequent DPCA processing. The mask matrices can be calculated from $\bar{\boldsymbol{R}}_{\boldsymbol{d}}$, which can be expressed as

$$\begin{aligned} \boldsymbol{M_1} &= \mathbf{1} - I_c(\boldsymbol{M}_{\text{out}}), \quad \boldsymbol{M_2} = \mathbf{1} - I_d(\boldsymbol{M}_{\text{in}}) \\ \boldsymbol{M}_{\text{ask}} &= \boldsymbol{M_1} \cdot \boldsymbol{M_2} \end{aligned} \tag{5}$$

where $\boldsymbol{M}_{\text{out}}, \boldsymbol{M}_{\text{in}} \in \mathbb{R}^{N_r \times N_d}$ are binary matrices marking the positions of strong points outside and inside the clutter band. I_c and I_d represent morphology closing operation and connected domain detection, respectively. $\boldsymbol{M}_{\text{ask}}$ is the final mask matrix. All resolution units in $\bar{\boldsymbol{R}}_{\boldsymbol{d}}$ are sorted in descending order according to their amplitudes and the top N_T points are selected. Then $\boldsymbol{M}_{\text{out}}$ can be obtained by determining whether these points are outside the clutter band (likewise for $\boldsymbol{M}_{\text{in}}$). In detail, if a selected point is outside the clutter band, the value in $\boldsymbol{M}_{\text{out}}$ at its position is set as 1.

3) DPCA With Adaptive Optimization: The optimization problem shown in (1) can also be solved by iteration along the descending direction of the gradient. Therefore, an adaptive optimization algorithm is proposed to obtain better DPCA results where the target's SCNR is significantly improved. The flowchart is shown in Fig. 2.

First, a 2-D complex weight matrix $\boldsymbol{W} \in \mathbb{C}^{N_r \times N_d}$ is initialized, which can be written by

$$\boldsymbol{W}(f_r, f_a) = \boldsymbol{H}(f_r)\boldsymbol{D}(f_a) \tag{6}$$

where $\boldsymbol{H} \in \mathbb{C}^{N_r \times 1}$ and $\boldsymbol{D} \in \mathbb{C}^{1 \times N_d}$ are learnable complex vectors which are used to further compensate the channel mismatch along fast-time and slow-time frequencies.

Second, the masked precalibrated return matrix $\bar{\boldsymbol{S}}_{\boldsymbol{c}}'$ is filtered by masked weight $\boldsymbol{W}'$, which can be expressed as

$$\begin{aligned} \hat{\boldsymbol{S}}_{\boldsymbol{c}} &= \bar{\boldsymbol{S}}_{\boldsymbol{c}}' \cdot \boldsymbol{W}' \\ \bar{\boldsymbol{S}}_{\boldsymbol{c}}' &= \bar{\boldsymbol{S}}_{\boldsymbol{c}} \cdot \boldsymbol{M}_{\text{out}} \\ \boldsymbol{W}' &= \boldsymbol{W} \cdot \boldsymbol{M}_{\text{ask}} + \mathbf{1} - \boldsymbol{M}_{\text{ask}} \end{aligned} \tag{7}$$

where $\hat{\boldsymbol{S}}_{\boldsymbol{c}}$ is the filtered return matrix. The purpose of mask matrices is to produce different filtering effects for strong points outside or inside the clutter band.

Third, the amplitude DPCA result is used as the main optimization object and the loss function can be written by

$$\text{loss}_a = \frac{1}{N_r N_d} \sum_{i=1}^{N_r} \sum_{j=1}^{N_d} \left\{ \left[\left| \hat{\boldsymbol{S}}_{\boldsymbol{c}}(f_{ri}, f_{aj}) \right| - \left| \boldsymbol{S_r}(f_{ri}, f_{aj}) \right| \right]^2 \times \boldsymbol{M}_{\text{ask}}(f_{ri}, f_{aj}) \right\}. \tag{8}$$

Equation (8) is the masked mean square error function and the purpose of $\boldsymbol{M}_{\text{ask}}$ is to ensure that only the nonpotential targets are suppressed. However, only the amplitude loss function is not enough since it cannot guarantee the correlation between $\hat{\boldsymbol{S}}_{\boldsymbol{c}}$ and $\boldsymbol{S_r}$. Therefore, a loss function of complex correlation coefficient is used, which can be expressed as

$$\begin{aligned} \text{loss}_c = 1 - \frac{1}{N_d} & \times \sum_{j=1}^{N_d} \left\{ \left| \sum_{i=1}^{N_r} \hat{\boldsymbol{E}}_{\boldsymbol{c}}(f_{ri}, t_{aj}) \boldsymbol{E_r}(f_{ri}, t_{aj})^* \right| \right. \\ & \left. \times \Big/ \sqrt{\sum_{i=1}^{N_r} |\hat{\boldsymbol{E}}_{\boldsymbol{c}}(f_{ri}, t_{aj})|^2 \sum_{i=1}^{N_r} |\boldsymbol{E_r}(f_{ri}, t_{aj})|^2} \right\} \end{aligned} \tag{9}$$

where $\hat{\boldsymbol{E}}_{\boldsymbol{c}}$ is the azimuth time-domain return matrix that can be obtained by inverse Fourier transform of $\hat{\boldsymbol{S}}_{\boldsymbol{c}}$ along the slow-time frequency dimensional (likewise for $\boldsymbol{E_r}$). Therefore, the total loss function can be written by

$$\text{Loss} = A\text{loss}_a + B\text{loss}_c \tag{10}$$

where A and B are balance coefficients. If the iteration step number does not reach the set value, the learnable vectors $\boldsymbol{H}$ and $\boldsymbol{D}$ are updated by an adaptive moment estimation (Adam) optimizer using the TensorFlow toolbox; otherwise, the iteration stops and the suppressed RD spectrum $\tilde{\boldsymbol{R}}_{\boldsymbol{d}}$ can be obtained by amplitude DPCA, where $\tilde{\boldsymbol{R}}_{\boldsymbol{d}} = |\hat{\boldsymbol{S}}_{\boldsymbol{c}}| - |\boldsymbol{S_r}|$.

Though the amplitude DPCA result is used, the main clutter and noise are effectively suppressed and the moving target's energy is protected well, thanks to the mask strategy and adaptive optimization.

III. Dual-DP-TBD Algorithm for Joint Detection

We propose a new Dual-DP-TBD algorithm to remove the limitations of moving target shadow detection by the conventional TBD algorithm, which embeds the Doppler supervision into an improved DP-TBD framework. The detailed flowchart is shown in Fig. 3.

In the preprocessing, sequential high-resolution SAR images are focused by an iterative backprojection algorithm (BPA) with autoregressive technique [30] and filtered by the mean filtering to suppress speckle noise. The imaging pulses are extracted from the reference channel. Meanwhile, partial consecutive pulses are extracted from the aligned reference and calibration channels around the center of each imaging

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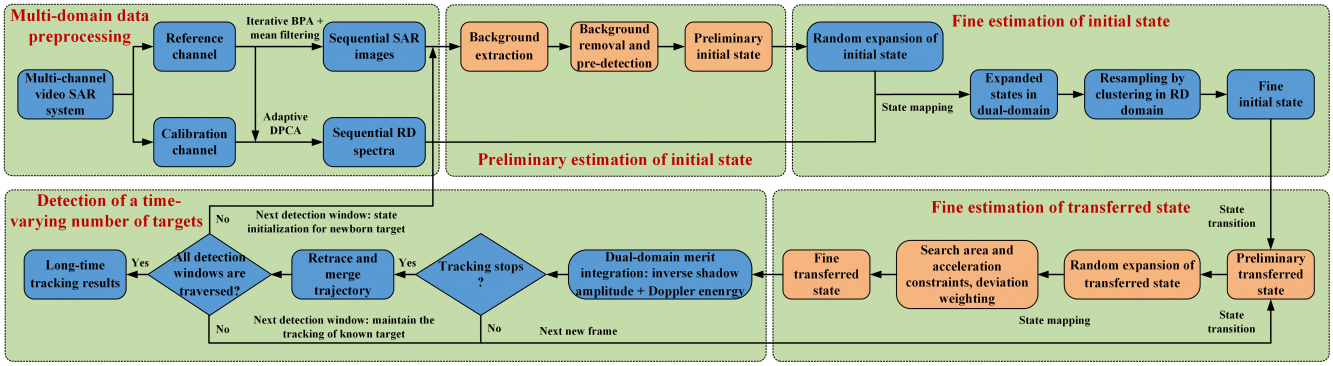


Fig. 3. Framework of the proposed Dual-DP-TBD algorithm for MF joint detection.

aperture, respectively. Subsequently, sequential low-resolution RD spectra are obtained by the proposed ADPCA method.

The classical shadow detection procedure is used for preliminary estimation of the initial state. However, there are large estimation errors, especially for the velocity components. Therefore, Doppler is used to provide effective velocity guidance and to realize fine estimation. The fine states can be searched in dual-domain through resampling from random expansion in state initialization and transition. This new state transition method does not depend on the traditional quantified acceleration process noises. The inverse shadow amplitude and Doppler energy are used as the dual-domain merit function which is integrated based on these fine states. Detection and tracking of newborn, known, and early-exiting targets are realized by using a time-varying detection threshold. The implementations of the framework are detailed below.

Before discussion, we give some symbol definitions. $\boldsymbol{\varphi}$ denotes the preliminary initial or transferred state. $\bar{\boldsymbol{\varphi}}$ is the expanded state. $\hat{\boldsymbol{\varphi}}$ denotes the fine initial or transferred state. $\tilde{\boldsymbol{\varphi}}$ is the declared state in retraced trajectory.

A. Foundation of DP-TBD for Shadow Detection

The shadow-based conventional SW-DP-TBD algorithm is briefly introduced. Measurements are sequential video SAR imagery with the sizes of $N_x \times N_y$. To realize the detection of a new target in time, the DP-TBD algorithm usually uses the SW strategy in the long-time tracking. Assume that there are K frames in continuous surveillance, and thus pixelated measurement model of the ℓth detection window can be expressed as

$$\boldsymbol{\varpi}_{1:N}^{\ell} = \left\{\boldsymbol{\alpha}_{1+(\ell-1)S}, \ldots, \boldsymbol{\alpha}_{n+(\ell-1)S}, \ldots, \boldsymbol{\alpha}_{N+(\ell-1)S}\right\} \quad (11)$$

where N and S are the length and step of the SW, respectively. $\boldsymbol{\alpha}_{n+(\ell-1)S}$ is the nth frame video SAR image in the ℓth detection window that can also be denoted as $\boldsymbol{\alpha}_n^{\ell}$.

Based on the 2-D pixelated measurement model, the state vector of the mth candidate target in $\boldsymbol{\alpha}_n^{\ell}$ can be expressed as

$$\boldsymbol{\varphi}_n^{\ell m} = \left[x_n^{\ell m} \quad \dot{x}_n^{\ell m} \quad y_n^{\ell m} \quad \dot{y}_n^{\ell m}\right]^T \quad (12)$$

where $x_n^{\ell m}$ and $y_n^{\ell m}$ are pixel coordinates, $\dot{x}_n^{\ell m}$ and $\dot{y}_n^{\ell m}$ are corresponding velocity components whose values are the same as the coordinate differences of this target in adjacent two frames. Symbol $\cdot^T$ is transpose operation.

Assume that target's motion follows a discrete nearly constant velocity (NCV) model and state transition satisfies linear Markov model, and thus the target dynamic model in the discrete state space can be expressed as [27]

$$\boldsymbol{\varphi}_n^{\ell m} = \boldsymbol{F}\boldsymbol{\varphi}_{n-1}^{\ell m} + \left[\delta\dot{x} \quad \delta\dot{y}\right]^T \otimes \left[1 \quad 1\right]^T \quad (13)$$

where $\delta\dot{x}$ and $\delta\dot{y}$ are quantified acceleration process noises. Symbol $\otimes$ denotes the Kronecker product. $\boldsymbol{F}$ is the state transition matrix, which can be expressed as

$$\boldsymbol{F} = \boldsymbol{I_2} \otimes \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \quad (14)$$

where $\boldsymbol{I_2}$ is a second-order identity matrix and t is the time interval between adjacent two frames. N_s valid transition states can be obtained by (13), where N_s depends on the number of used $\delta\dot{x}$ and $\delta\dot{y}$.

For well-separated targets, multitarget tracking can be solved by implementing several single-target DP integrations [23]. Therefore, the merit integration can be expressed as

$$\begin{aligned} \Theta\left(\boldsymbol{\varphi}_n^{\ell m}\right) &= \max_{\boldsymbol{\varphi}_{n-1}^{\ell m} \in \boldsymbol{\tau}\left(\boldsymbol{\varphi}_n^{\ell m}\right)} \left[\Theta\left(\boldsymbol{\varphi}_{n-1}^{\ell m}\right)\right] + \boldsymbol{v}_n^{\ell}\left(\boldsymbol{\varphi}_n^{\ell m}\right) \\ \boldsymbol{\Psi}_n^{\ell m}\left(\boldsymbol{\varphi}_n^{\ell m}\right) &= \arg \max_{\boldsymbol{\varphi}_{n-1}^{\ell m} \in \boldsymbol{\tau}\left(\boldsymbol{\varphi}_n^{\ell m}\right)} \left[\Theta\left(\boldsymbol{\varphi}_{n-1}^{\ell m}\right)\right] \end{aligned} \quad (15)$$

where $\Theta(\boldsymbol{\varphi}_n^{\ell m})$ is integrated merit of state $\boldsymbol{\varphi}_n^{\ell m}$, $\boldsymbol{\tau}(\boldsymbol{\varphi}_n^{\ell m})$ is a set of transferred states which can transfer to $\boldsymbol{\varphi}_n^{\ell m}$ at discrete time $n-1$. $\boldsymbol{v}_n^{\ell}(\boldsymbol{\varphi}_n^{\ell m})$ is the inverse pixel value at the position of $\boldsymbol{\varphi}_n^{\ell m}$, that is, $\boldsymbol{v}_n^{\ell} = 255 - \boldsymbol{\alpha}_n^{\ell}$. $\boldsymbol{\Psi}_n^{\ell m}(\cdot)$ is a retracing function.

In the last frame of the ℓth detection window, the final decision of a single-target can be expressed as [27]

$$\begin{aligned} \tilde{\boldsymbol{\varphi}}_N^{\ell m} &= \arg \max_{\boldsymbol{\varphi}_N^{\ell m} \in \boldsymbol{\Lambda}_N^{\ell m}} \left[\Theta\left(\boldsymbol{\varphi}_N^{\ell m}\right)\right] \\ &\text{s.t. } \Theta\left(\tilde{\boldsymbol{\varphi}}_N^{\ell m}\right) > V_T \end{aligned} \quad (16)$$

where V_T is the detection threshold, and $\boldsymbol{\Lambda}_N^{\ell m}$ and $\tilde{\boldsymbol{\varphi}}_N^{\ell m}$ are the set of all transferred states and declared state in the Nth frame, respectively. The declared trajectory can be obtained by exploiting the retracing function.

B. Preliminary Estimation of Initial State

The computational load of the DP-TBD algorithm will be huge if it directly operates on the original measurement.

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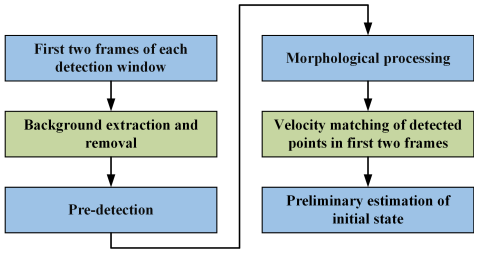


Fig. 4. Classical shadow detection procedure for state initialization.

Therefore, a predetection is needed to reduce the number of candidate targets. Here, a classical shadow detection procedure is used for state initialization, which is shown in Fig. 4.

The classical shadow detection is implemented in the first two frames $\{\boldsymbol{\alpha}_n^\ell\}_{n=1,2}$ of each detection window. Before predetection, background removal is used to suppress stationary clutter and to highlight the moving target shadow in the SAR image. Since the SAR image is focused by iterative BPA, stationary clutter is almost located at the same resolution units in adjacent frames. Therefore, the backgrounds $\{\boldsymbol{\beta}_n^\ell\}_{n=1,2}$ can be simply extracted by calculating the mean value of the reference image and its following three frames. Then, the clutter can be suppressed by removing the backgrounds, and the moving target shadow is distinguished in the residual images $\{\boldsymbol{\alpha}_n^\ell/\boldsymbol{\beta}_n^\ell\}_{n=1,2}$. A constant threshold and morphological processing, such as closing, opening operations, and connected domain detection, are used to detect shadows. The centers of connected components are considered the detected points.

A simple velocity matching method is used for state initialization. Assume that $(x_1^{\ell m}, y_1^{\ell m})$ and $(x_2^{\ell j}, y_2^{\ell j})$ are detected points in the first and second frames of the ℓth detection window, respectively. If the differences of pixel coordinates (velocities between frames) satisfy $\Delta x \in [\dot{x}_{\min}, \dot{x}_{\max}]$ and $\Delta y \in [\dot{y}_{\min}, \dot{y}_{\max}]$, detected point $(x_1^{\ell m}, y_1^{\ell m})$ can be regarded as a candidate target and its initial state can be expressed as

$$\boldsymbol{\varphi}_1^{\ell m} = \begin{bmatrix} x_1^{\ell m} & \Delta x & y_1^{\ell m} & \Delta y \end{bmatrix}^T \tag{17}$$

where $\Delta x = x_2^{\ell j} - x_1^{\ell m}$ and $\Delta y = y_2^{\ell j} - y_1^{\ell m}$. The initial state set can be obtained by traversing all detected points in the first frame and matching all detected points in the second frame according to the above mentioned method.

C. Fine Estimation of Initial State

In preliminary estimation, the shadow position is considered the center of the corresponding connected component. Since the boundary of the moving target shadow is usually smeared, it is very difficult to accurately locate the shadow. After velocity matching, this error will be included in the estimated velocity, which brings a heavy influence on target tracking.

Fortunately, Doppler can be used to correct the preliminary state. First, we give definitions of the same state in different representations. In the SAR image, a state vector can be defined in a pixelated measurement model shown in (12), here, we rewrite it as $\boldsymbol{\varphi}_{nP}^{\ell m}$. Meanwhile, it can also be defined in imaging coordinate system that uses the geographic coordinate. The state can be expressed as $\boldsymbol{\varphi}_{nI}^{\ell m}$ and the units of its position and velocity are m and m/s, respectively. Note

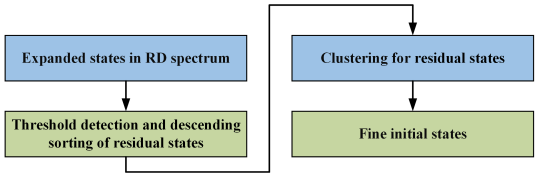


Fig. 5. Resampling of fine initial states from expanded states.

that $\boldsymbol{\varphi}_{nP}^{\ell m}$ and $\boldsymbol{\varphi}_{nI}^{\ell m}$ are equivalent representations and can be converted to each other according to imaging grid. Assume that $\boldsymbol{\kappa}_n^\ell$ is the corresponding RD spectrum to the SAR image $\boldsymbol{\alpha}_n^\ell$, the coordinate $(r_n^{\ell m}, f_n^{\ell m})$ of state $\boldsymbol{\varphi}_{nI}^{\ell m}$ in RD domain can be expressed as

$$\begin{aligned} r_n^{\ell m} &= \left\langle \frac{N_r'}{2} + \frac{(R_s - R_t) + \Delta r}{\Delta R} \right\rangle, \quad \Delta R = \frac{c F_s}{2\gamma N_r'} \\ f_n^{\ell m} &= \mathbb{F}\left[\frac{2(v_{rr} + v_{rt})}{\lambda}\right] \end{aligned} \tag{18}$$

where N_r' is the number of fast-time samples of original dechirp returns, R_s is the reference distance, R_t is the Euclidean distance between the positions of state $\boldsymbol{\varphi}_{nI}^{\ell m}$ and radar, Δr is a compensation term which compensates the range inconsistency between moving target's shadow and Doppler energy [29], c is the speed of light, F_s is the sampling frequency, γ is the fast-time frequency modulation rate, and $\langle\cdot\rangle$ represents rounding operation. v_{rr} and v_{rt} are the projections of radar velocity and state velocity in the direction of sight line, respectively. λ is the wavelength. Function $\mathbb{F}(\cdot)$ converts the Doppler frequency into Doppler index. State mapping from the SAR image to the RD spectrum can be realized by (18).

Based on state mapping, we propose a fine estimation of the initial state which significantly corrects the error in preliminary estimation. This fine estimation is implemented in three steps.

First, possible states are generated by expansion [27]. An expanded initial state set $\bar{\boldsymbol{\Omega}}_{1I}^{\ell m}$ corresponding to preliminary initial state $\boldsymbol{\varphi}_{1I}^{\ell m}$ is sampled from random distributions, which can be expressed as

$$\begin{aligned} \boldsymbol{X}_P &= x_{nP}^{\ell m} + \boldsymbol{U}_D(x_a, x_b) \\ \boldsymbol{Y}_P &= y_{nP}^{\ell m} + \boldsymbol{U}_D(y_a, y_b) \\ \dot{\boldsymbol{X}}_I &= \boldsymbol{N}_D\left(\dot{x}_{nI}^{\ell m}, \sigma_x\right) \\ \dot{\boldsymbol{Y}}_I &= \boldsymbol{N}_D\left(\dot{y}_{nI}^{\ell m}, \sigma_y\right) \end{aligned} \tag{19}$$

where $n = 1$ in the case of fine estimation of initial state, D is the number of samples, and $\boldsymbol{X}_P$ is expanded pixel coordinate set sampled from a uniform distribution $\boldsymbol{U}(x_a, x_b)$ around the original position $x_{nP}^{\ell m}$. $\dot{\boldsymbol{X}}_I$ is expanded real velocity set sampled from a Gaussian distribution with mean value $\dot{x}_{nI}^{\ell m}$ and standard deviation σ_x. Moreover, $\boldsymbol{X}_P$ is converted to $\boldsymbol{X}_I$ (likewise for $\boldsymbol{Y}_P$ and $\dot{\boldsymbol{Y}}_I$). In imaging geometry, x- and y-axes represent the azimuth and range directions, respectively.

Second, each state in $\bar{\boldsymbol{\Omega}}_{1I}^{\ell m} = \{\bar{\boldsymbol{\varphi}}_d\}_{d=1}^D$ is mapped into the RD spectrum $\boldsymbol{\kappa}_1^\ell$. Its coordinate (r_d, f_d) is calculated by (18).

Third, the fine initial states are resampled from expanded states. The procedure can be seen in Fig. 5. A threshold detection is used to screen expanded states in the RD spectrum. If the RD value $\boldsymbol{\kappa}_1^\ell(r_d, f_d)$ is higher than threshold V_1, this state is retained; otherwise, it is deleted as invalid state.

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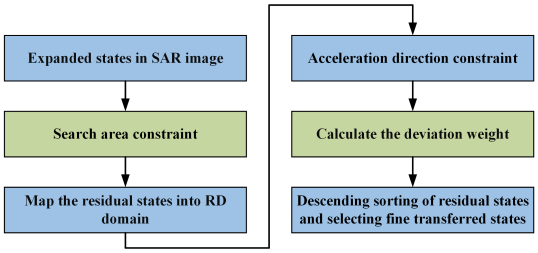


Fig. 6. Resampling of fine transferred states from expanded states.

Then, D_1 residual states are sorted in descending order according to their RD values, and a simple clustering algorithm is used to screen the fine initial states. The algorithm flowchart is shown in Algorithm 1, where V_2 and V_3 are distance and mean value thresholds, respectively. The purpose of neighborhood mean value is to avoid the interferences of isolated strong points in the RD spectrum. Note that a preliminary initial state $\boldsymbol{\varphi}_{1I}^{\ell m}$ may correspond to multiple fine initial states $\hat{\boldsymbol{\Omega}}_{1I}^{\ell m} = \{\hat{\boldsymbol{\varphi}}_p\}_{p=1}^{P}$, where P is the total number of fine initial states.

Algorithm 1 Clustering in RD Domain for Screening Fine Initial States

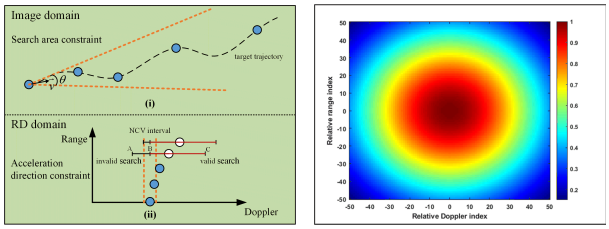
Input: Descending residual states $\{\bar{\boldsymbol{\varphi}}_d\}_{d=1}^{D_1}$.
Output: Fine initial states.
1: Initialization: cluster set $\boldsymbol{\varepsilon} = \{\bar{\boldsymbol{\varphi}}_1\}$.
2: **for** $d = 2$ to D_1 **do**
3: Calculate the 2-dimensional Euclidean distances $\boldsymbol{R}$ between state $\bar{\boldsymbol{\varphi}}_d$ and all states in cluster set $\boldsymbol{\varepsilon}$ in RD domain.
4: Calculate the neighborhood mean value ϵ of state $\bar{\boldsymbol{\varphi}}_d$ in RD domain.
5: **if** $\boldsymbol{R} > V_2$ and $\epsilon > V_3$ **then** add state $\bar{\boldsymbol{\varphi}}_d$ into cluster set $\boldsymbol{\varepsilon}$.
6: **end if**
7: **end for**
8: All states in cluster set $\boldsymbol{\varepsilon}$ are considered as fine initial states.

The fine estimation of the initial state is essentially a resampling of the preliminary estimate, which compensates for the error introduced by predetection and is helpful for maneuvering target tracking.

D. Fine Estimation of Transferred State

Traditional state transition shown in (13) relies on quantified acceleration process noises to track the maneuvering target. The number of valid transition states N_s has a great influence on the detection and tracking performance. The small N_s is not enough to track the maneuvering target, while the large N_s makes the predicted state space rapidly expand with the increase of the number of integrated frames, which will cover the other weak reflectivity areas in SAR images and increase the computational load. Furthermore, the quantified acceleration process noises are usually set as symmetrical discrete sequences to give consideration to the possible moving direction and maneuverability. As a result, the predicted state space presents a regular and dense rectangular distribution that contains a large number of invalid states.

To overcome these problems, a new state transition is proposed without the quantified acceleration process noises. Assume that a fine transferred state in discrete time $n-1$ is $\hat{\boldsymbol{\varphi}}_{(n-1)I}^{\ell m}$. The preliminary state transition can be



(a) (b)

Fig. 7. Illustration of some procedures in resampling of fine transferred states. (a) Schematic of the search area and acceleration direction constraints. The blue circles represent fine initial or transferred states and the white circles are preliminary transferred states. (b) Visualization of deviation weight.

expressed as

$$\boldsymbol{\varphi}_{nI}^{\ell m} = \boldsymbol{F}\hat{\boldsymbol{\varphi}}_{(n-1)I}^{\ell m} \tag{20}$$

where $\boldsymbol{\varphi}_{nI}^{\ell m}$ is the preliminary transferred state. However, only the constant velocity model cannot deal with the maneuvering target tracking. Hence, a resampling is also needed to generate the fine transferred states, which is implemented in two steps.

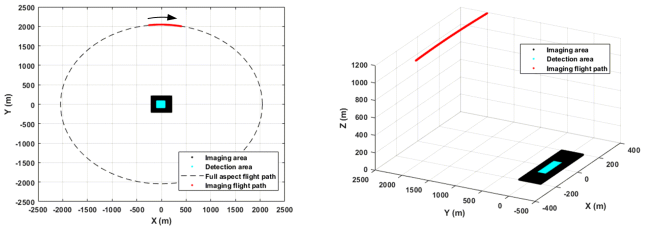
First, an expanded transferred state set $\bar{\boldsymbol{\Omega}}_{nI}^{\ell m} = \{\bar{\boldsymbol{\varphi}}_j\}_{j=1}^{J}$ corresponding to $\boldsymbol{\varphi}_{nI}^{\ell m}$ is obtained using the random expansion shown in (19), where J is the number of expanded states. Second, N_s' fine transferred states are selected from $\bar{\boldsymbol{\Omega}}_{nI}^{\ell m}$. Fig. 6 shows the flowchart of the resampling, detailed as follows.

1) Search Area Constraint: The expanded states in the SAR image are first screened based on the hypothesis that the real trajectory of a moving target in a detection window is located in a fan-shaped area with the intersection angle of θ and the unit velocity vector $\vec{\boldsymbol{v}}$ of the fine initial state $\hat{\boldsymbol{\varphi}}_{1I}^{\ell m}$ is the angular bisector, as shown in Fig. 7(a). The hypothesis is true when the length of the detection window N is small. The expanded states outside the fan-shaped area are deleted and the residual states are mapped into the RD domain.

2) Acceleration Direction Constraint: After the third frame, the motion model, that is, NCV, acceleration, or deceleration, of a moving target can be determined by the variation of its Doppler. Assume that the motion model is unchanged in a short detection window, and thus the expansion becomes directional in the Doppler dimension. Therefore, the expanded states that do not satisfy the motion model are deleted. As shown in Fig. 7(a), interval AC is the distribution of all expanded states in the fourth frame and AB is considered an invalid search interval since this target is accelerated.

3) Calculate the Deviation Weight: The large σ_x and σ_y in (19) will produce a large search area along the Doppler dimension, so that the expanded states deviated from preliminary transferred state $\boldsymbol{\varphi}_{nI}^{\ell m}$ can cover the real position of moving target. However, the resampling may be wrong and leads to incorrect tracking due to the interference of residual clutter and dense energies of multiple targets if the expanded states with different deviation degrees are equivalent. Therefore, a Gaussian weight w centered at the RD position of $\boldsymbol{\varphi}_{nI}^{\ell m}$ is used to avoid searching for incorrect strong point. The visualization of w can be seen in Fig. 7(b), which can be

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(a) (b)

Fig. 8. Imaging geometry. (a) and (b) 2-D and 3-D illustrations, respectively. The movement direction of UAV is indicated by the black arrow. The imaging area, detection area, and imaging flight path are indicated by black, cyan, and red points, respectively.

expressed as

$$w(\bar{\boldsymbol{\varphi}}_j) = \exp\left[-\frac{(r_j - r_c)^2 + (f_j - f_c)^2}{2\sigma^2}\right] \tag{21}$$

where (r_j, f_j) and (r_c, f_c) are the RD coordinates of expanded state $\bar{\boldsymbol{\varphi}}_j$ and preliminary transferred state $\boldsymbol{\varphi}_{nI}^{\ell m}$, respectively.

4) Resampling: The residual expanded states after constraints and threshold detection are sorted in descending order according to their weighted RD values. The top N_s' states are selected as the fine transferred states $\hat{\boldsymbol{\Omega}}_{nI}^{\ell m} = \{\hat{\boldsymbol{\varphi}}_i\}_{i=1}^{N_s'}$.

The proposed state transition approach can accurately search moving targets in dual-domain due to resampling and Doppler supervision. The fine transferred states are used to integrate the merit function in the DP process, which can significantly improve the tracking performance of the maneuvering target.

E. Detection of a Time-Varying Number of Targets

Different from traditional wide-area surveillance, video SAR is usually used for the small-size scene, and thus there is a more time-varying number of targets. In detail, targets in video SAR images can be divided into three categories.

1) Newborn Target: If an initialized candidate target is not detected in the last window, this target is called a newborn target and will be tracked over N frames in the current detection window. Sections III-B–III-D show the complete procedures for the newborn target.

2) Known Target: If a target's trajectory has been declared in the last detection window, this target is called a known target and will be continuously tracked from the end of the last window to the end of the current window. In particular, the fine initial state is directly inherited from the declared state at the end of the last detection window. Therefore, the state initialization is avoidable. The known target only needs to be tracked over $S+1$ frames by using the recursive implementation [25].

3) Early-Exiting Target: In the tracking process, if a target exits the observation scene without being integrated over the specified number of frames (N frames for the newborn target and $S+1$ frames for the known target), this target is called an early-exiting target. Assume that all preliminary transferred states of a target are out-of-bounds in the $(\eta+1)$th frame, this target exits the observation scene at the discrete-time $(\eta+1)$ and its merit function integration ends at the discrete-time η.

The merit function is very important for DP-TBD. The moving target shadow in the SAR image and its Doppler energy in the RD spectrum are typical features. Both of them can be used to distinguish the moving target from stationary clutter. However, robust tracking cannot be realized by only relying on either shadow or Doppler energy. On the one hand, there are many weak reflectivity areas in the SAR image that have the similar amplitude level to the moving target shadow. The speckle noise also deteriorates the merit integration. On the other hand, it is difficult to distinguish moving target from strong interferences in the RD spectrum, such as residual clutter or noise, especially for the slow target. Fortunately, a real moving target owns both the shadow and Doppler energy. Therefore, robust moving target detection and tracking can be realized by simultaneously utilizing the shadow and Doppler energy, the dual-domain merit function used for joint detection can be expressed as

$$\begin{aligned}\Theta_{\text{dual}}(\hat{\boldsymbol{\varphi}}_n^{\ell m}) = &\max_{\hat{\boldsymbol{\varphi}}_{n-1}^{\ell m} \in \boldsymbol{\tau}(\hat{\boldsymbol{\varphi}}_n^{\ell m})} \left[\Theta_{\text{dual}}(\hat{\boldsymbol{\varphi}}_{n-1}^{\ell m})\right] \\ &+ w(\hat{\boldsymbol{\varphi}}_n^{\ell m}) \boldsymbol{\nu}_n^{\ell}(\hat{\boldsymbol{\varphi}}_n^{\ell m}) + w(\hat{\boldsymbol{\varphi}}_n^{\ell m}) \boldsymbol{\kappa}_n^{\ell}(\hat{\boldsymbol{\varphi}}_n^{\ell m})\end{aligned} \tag{22}$$

where the second and third terms are the weighted inverse shadow amplitude and Doppler energy, respectively. The local optimal state in the discrete time $n-1$ which can transfer to state $\hat{\boldsymbol{\varphi}}_n^{\ell m}$ is written by

$$\dot{\boldsymbol{\varphi}}_{n-1}^{\ell m} = \arg \max_{\hat{\boldsymbol{\varphi}}_{n-1}^{\ell m} \in \boldsymbol{\tau}(\hat{\boldsymbol{\varphi}}_n^{\ell m})} \left[\Theta_{\text{dual}}(\hat{\boldsymbol{\varphi}}_{n-1}^{\ell m})\right]. \tag{23}$$

During the DP process, the weighted inverse shadow amplitude and Doppler energy are also individually integrated to more specifically characterize the moving target, which can be expressed as

$$\begin{aligned}\Theta_{\text{Im}}(\hat{\boldsymbol{\varphi}}_n^{\ell m}) &= \Theta_{\text{Im}}\left(\dot{\boldsymbol{\varphi}}_{n-1}^{\ell m}\right) + w(\hat{\boldsymbol{\varphi}}_n^{\ell m}) \boldsymbol{\nu}_n^{\ell}(\hat{\boldsymbol{\varphi}}_n^{\ell m}) \\ \Theta_{\text{Rd}}(\hat{\boldsymbol{\varphi}}_n^{\ell m}) &= \Theta_{\text{Rd}}\left(\dot{\boldsymbol{\varphi}}_{n-1}^{\ell m}\right) + w(\hat{\boldsymbol{\varphi}}_n^{\ell m}) \boldsymbol{\kappa}_n^{\ell}(\hat{\boldsymbol{\varphi}}_n^{\ell m}).\end{aligned} \tag{24}$$

Therefore, the joint detection of the proposed Dual-DP-TBD algorithm in the ℓth detection window can be expressed as

$$\begin{aligned}\tilde{\boldsymbol{\varphi}}_{\mu}^{\ell m} = &\arg \max_{\hat{\boldsymbol{\varphi}}_{\mu}^{\ell m} \in \boldsymbol{\Lambda}_{\mu}^{\ell m}} \left[\Theta_{\text{dual}}(\hat{\boldsymbol{\varphi}}_{\mu}^{\ell m})\right] \\ &\text{s.t. } \Theta_{\text{Im}}(\tilde{\boldsymbol{\varphi}}_{\mu}^{\ell m}) > \mu V_{\text{Im}} \Theta_{\text{Rd}}(\tilde{\boldsymbol{\varphi}}_{\mu}^{\ell m}) > \mu V_{\text{Rd}}\end{aligned} \tag{25}$$

where μ represents the integrated number of frames, that is, μ equals to N, $S+1$, and η for newborn, known, and early-exiting targets, respectively. $\boldsymbol{\Lambda}_{\mu}^{\ell m}$ is the set of all fine transferred states in the μth frame. V_{Im} and V_{Rd} are detection thresholds for inverse shadow amplitude and Doppler energy, respectively. $\tilde{\boldsymbol{\varphi}}_{\mu}^{\ell m}$ is the declared state and the declared trajectory can be obtained by exploiting the retracing function.

In the long-time tracking, a short trajectory obtained in a single detection window is merged into the previous trajectory of the same target to generate a longer trajectory. The time-varying detection threshold shown in (25) can handle the detection and tracking of a time-varying number of targets well. Moreover, the dual-domain merit function improves the robustness of moving target detection in video SAR.

In a single detection window, the computational complexity of the proposed Dual-DP-TBD algorithm mainly depends on

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TABLE I
PARAMETERS OF W-BAND UAV-BORNE VIDEO SAR SYSTEM

Parameter	Value	Parameter	Value
Center Frequency	94 GHz	Bandwidth	1 GHz
Pulse Repetition Interval	60 μs	Pulse Duration	12 μs
Sampling Frequency	240 MHz	Platform Velocity	42 m/s
Reference Distance	2.4 Km	Platform Height	1.2 Km

the parameters D, J, and N_s', where D and J are the number of expanded states in state initialization and transition, N_s' is the number of valid transition states of our approach. To track the maneuvering target accurately, the conventional DP-TBD usually uses a large N_s. However, the proposed algorithm can use an extremely small N_s' since the used transferred states are fine, for example, $N_s' = 3$ in our experiment. Therefore, the proposed Dual-DP-TBD algorithm has a lower computational load than the conventional DP-TBD algorithm under the appropriate D and J. In the subsequent experiment, we will give a detailed comparison of computational load.

IV. RESULTS OF VIDEO SAR MEASURED DATA

Both the ADPCA and Dual-DP-TBD algorithms have been verified by the measured data of a four-channel UAV-borne video SAR operated at W-band with circular spotlight mode [3]. The typical parameters are illustrated in Table I.

For a small-size scene, the classical dechirp technology is used to reduce the sampling bandwidth. The returns in the reference channel are focused by the iterative BPA with autoregressive technique [30] on the ground-range plane. The theoretical range resolution is 0.15 m on the slant-range plane, while it is about 0.17 m on the ground-range plane. The azimuth integration angle of each subaperture is approximately 0.48°, leading to a theoretical azimuth resolution of 0.19 m. $K = 33$ successive nonoverlapping SAR images are obtained and the total observation time is about 15.84 s. The imaging geometry is introduced in Fig. 8. It can be calculated that the aspect angle of Frame 1 is 95.84°, while it is 80.40° in Frame 33. The size of the whole imaging area on the ground is 400 m × 400 m. The detection area mainly focuses on a small area near the intersection which is a hot spot that contains multiple maneuvering targets with different states (accelerating, decelerating, turning, and stopping), and thus the SAR image with the pixel size of 1024 × 1024 is cropped from the original image for detection and tracking. Therefore, the size of the detection area is 153.54 m × 153.54 m. Another adjacent azimuth channel is considered the calibration channel. After aligning returns, $N_d = 1024$ consecutive pulses are selected around the center of each imaging aperture to obtain the original RD spectra of reference and calibration channels. Since the coherent processing time is only 0.06 s, the energies of a scatterer are located at the same resolution unit in two RD spectra. In preprocessing, the suppressed RD spectra are obtained by using the proposed ADPCA. Moreover, the classical DPCA and DPCA after CE (CE + DPCA) [15] are used for comparison. Then, the Dual-DP-TBD algorithm is implemented to detect the maneuvering target. The conventional SW-DP-TBD and R-MF-TBD algorithms [25] are examined for comparison.

A. Results of Clutter Suppression

The results of clutter suppression are shown in Fig. 9. For intuitive comparison, these spectra are normalized by the maximum of the suppressed RD spectrum obtained by DPCA. Fig. 10(a) shows the comparison of normalized intensity sums along the Doppler dimension and Fig. 10(b) shows the comparison of SCNRs of typical targets which are calculated in rectangular areas centered on the targets' center in RD spectrum. Note that the last four (three) moving targets in the left (right) subfigure in Fig. 10(b) are the targets within the clutter band.

In the reference RD spectrum, the resampling of fine state and Doppler energy integration will be deteriorated with the interferences of strong clutter and noise. Due to channel mismatch, the classical DPCA fails to suppress the stationary clutter. As shown in the second column of Fig. 9 and the green curves in Fig. 10(a), the intensities of clutter and noise are not reduced since their phases are not equal or even reverse in two channels. The SCNRs of direct DPCA processing shown in Fig. 10(b) may be higher or lower than that in the reference spectrum, which depends on the phases of corresponding resolution units between two channels. Conventional CE calibrates the amplitude and phase differences using the iteration in (2). It can be intuitively seen in the third column of Fig. 9 that both the clutter and target are suppressed after CE + DPCA processing compared to the reference spectrum while there is no obvious improvement for noise outside the clutter band. This phenomenon is also verified by blue curve in Fig. 10(a). The CE + DPCA processing has a weak ability of clutter suppression and denoise. Moreover, the inherent differences of moving targets are destroyed due to the entire integration, resulting in the loss of target energy during DPCA. The SCNR has no improvement for moving targets outside the clutter band since the noise in its neighborhood is almost the same.

The proposed ADPCA approach uses the preequalization strategy improved by partial summation and masked adaptive optimization based on gradient descent. In preequalization, the fast-time frequencies of top $F_N = 30$ strong points and all slow-time frequencies belonging to the clutter band are used. Subsequently, the top 7000 and 2000 points in the presuppressed spectrum $\bar{\boldsymbol{R}}_{\boldsymbol{d}}$ are selected to obtain the mask matrices $\boldsymbol{M}_{\text{out}}$ and $\boldsymbol{M}_{\text{in}}$, respectively. Finally, the optimization is implemented and the suppressed RD spectra are shown in the fourth column of Fig. 9 and the red curves in Fig. 10(a). It has the obvious improvement of SCNR, especially for a target outside the clutter band due to mask and denoise. The target's energy is protected and the mean SCNR improvement is about 3.5 dB for all targets. Moreover, both the clutter and noise are effectively suppressed. Hence, the proposed ADPCA is helpful for subsequent resampling of fine state and Doppler energy integration in the RD domain.

B. Results of Moving Target Detection

In the long-time tracking, $K = 33$ sequential video SAR images and corresponding suppressed RD spectra are divided into eight detection windows with the batch length $N = 5$ and step $S = 4$. The speckle noise in the SAR image is suppressed

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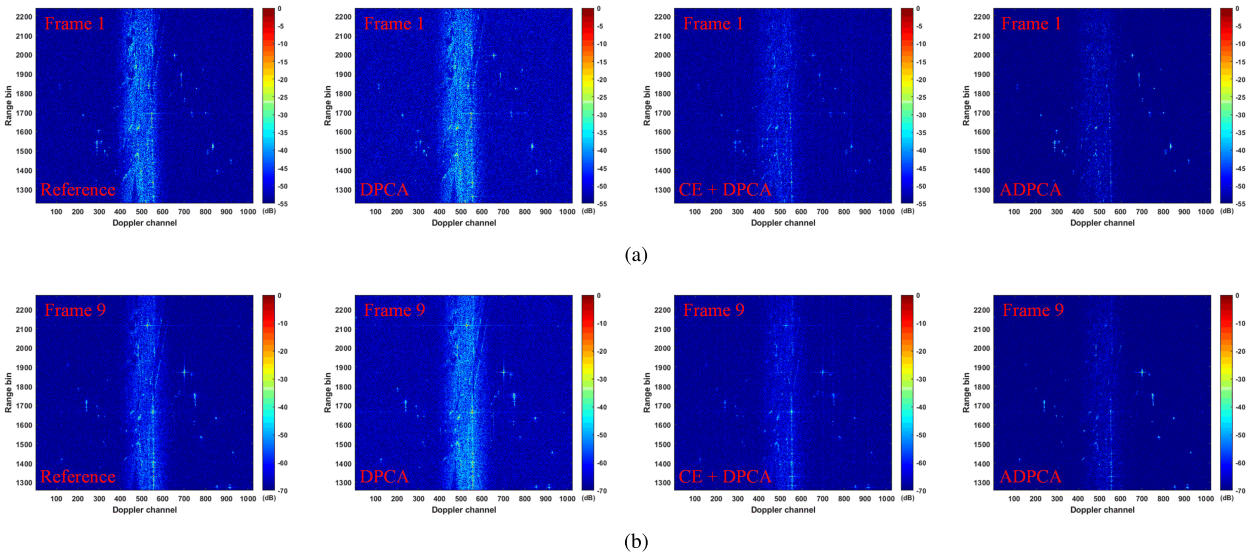


Fig. 9. Comparison of clutter suppression results. In each row, the reference RD spectrum and the suppressed RD spectra obtained by DPCA, DPCA after CE (CE + DPCA), and the proposed ADPCA are shown from left to right. (a) RD spectra of Frame 1. (b) RD spectra of Frame 9.

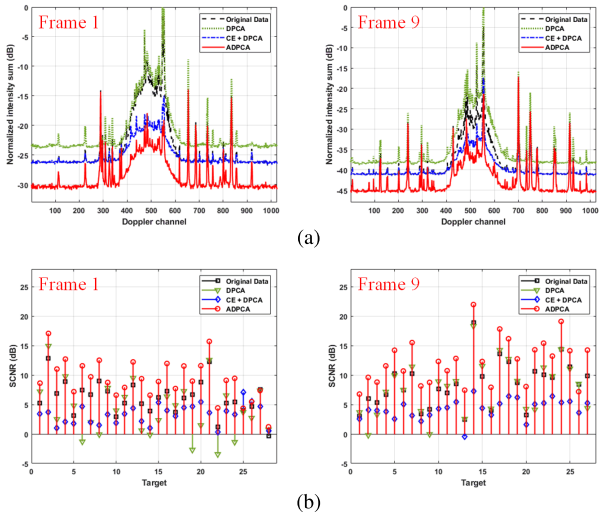


Fig. 10. Details of clutter suppression. In each row, the results of Frames 1 and 9 are shown from left to right. (a) Comparison of normalized intensity sums along the Doppler dimension. (b) Comparison of SCNRs of typical targets.

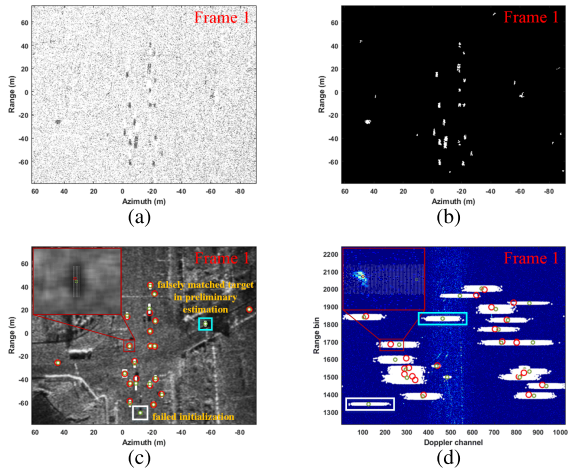


Fig. 11. State initialization for the first detection window. (a) Residual SAR image after background removal. (b) Predetection results. (c) and (d) Fine estimation of the initial state in the SAR image and the RD spectrum, respectively. The positions of the preliminary, expanded, and fine initial state are indicated by green circles, white points, and red circles, respectively.

by the mean filter. The amplitudes of the suppressed RD spectrum are normalized by their maximum and converted to decibels.

Fig. 11 shows state initialization for the first detection window. The preliminary estimation of the initial state uses the classical shadow detection procedure. The residual SAR image after background removal of Frame 1 is shown in Fig. 11(a). A constant threshold that equals 140 is used to detect shadow. In detail, since the moving target shadow is the low-gray area in the residual image, the pixel whose value less than the given constant threshold is considered a potential shadow. The choice of constant threshold should ensure that the real moving target shadow can be detected. After morphological processing, the predetection results are shown in Fig. 11(b). A simple velocity matching method is used for preliminary estimation, and the positions of preliminary initial states are indicated by green circles in Fig. 11(c) and (d). It can be seen that the mapped states in the RD spectrum have large velocity estimation errors and their positions deviate from the targets' real Doppler energies. Therefore, a fine estimation is needed. The expansion parameters of position, such as y_a and y_b, are adaptively adjusted according to the target's size obtained in predetection. On the one hand, the choice of expansion parameters of velocity σ_x and σ_y should ensure that the expanded states, such as the white points in Fig. 11(d), can cover the target's real Doppler energy well. On the other hand, too large σ_x and σ_y may make the expanded states cover many interferences, such as the isolated strong noise and residual clutter, which should be avoided. Therefore, for the target whose principal velocity component falls in the range direction, $\sigma_x = 0.05$ and $\sigma_y = 0.8$. In contrast, σ_x and σ_y

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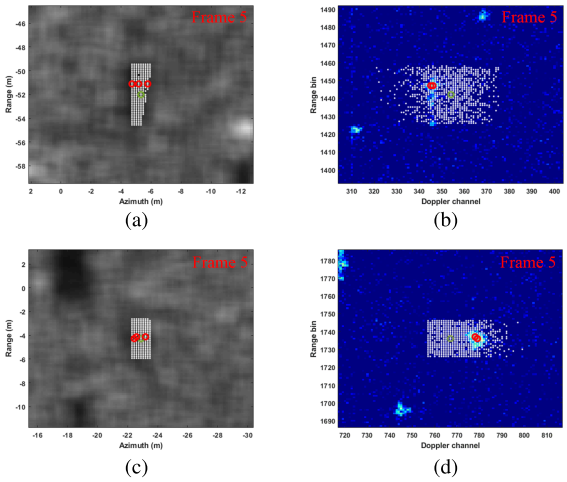


Fig. 12. State transition for the first detection window. Each subgraph shows a preliminary transferred state and its expanded and fine transferred states. (a) and (c) States in the SAR image while (b) and (d) in the RD spectrum. The positions of the preliminary, expanded, and fine transferred states are indicated by green circles, white points, and red circles, respectively.

equal to 1 and 0.1, respectively, if it falls in the azimuth direction. Finally, the fine initial states are resampled by threshold detection and clustering, which are indicated by red circles. The fine initial states accurately cover the positions of both the shadow and Doppler energy, indicating that there are negligible estimation errors. In addition, there may be falsely matched targets in preliminary estimation such as the target indicated by a cyan rectangle. However, it is not initialized by fine estimation since there is no Doppler energy to match the wrong velocity. Therefore, the fine estimation has the ability to suppress false alarms. There is a failed initialization indicated by the white rectangle due to the wrong velocity. It can be concluded that the fine estimation can compensate for the error introduced by predetection and provide an accurate initial state for maneuvering target tracking.

Two examples of state transition are given in Fig. 12. The preliminary transferred state indicated by the green circle cannot cover the target's real Doppler energy due to strong maneuverability. Since its velocity slightly deviates from the real velocity, the expansion parameters of velocity can be smaller than that set in state initialization. For example, σ_x and σ_y are equal to 0.05 and 0.3 for the target whose principal velocity component falls in the range direction, respectively. The residual expanded states after search area and acceleration direction constraints are indicated by white points. Finally, $N_s' = 3$ fine transferred states are selected. The proposed state transition approach can timely correct the preliminary transferred state to make it close to the real state of the moving target. It precisely searches the moving target in dual-domain due to resampling and Doppler supervision. Figs. 11 and 12 show that the second contribution of our work, that is, fine state estimation by using Doppler supervision, enables the joint detection in dual-domain. Therefore, it has the most beneficial impact on detection and tracking performance.

Fig. 13 gives the detailed comparison of the detection results. The first column shows the ground truths which mark the real positions of moving targets. The second and third columns show the results of the SW-DP-TBD and R-MF-TBD, respectively. The number of valid transition states N_s equals 36 and the initial states are also obtained by the classical shadow detection procedure. The fourth column shows the results of the proposed Dual-DP-TBD algorithm. There are a large number of maneuvering targets with different states during this observation period, which brings great difficulty to track. The good control ability to miss alarm of the proposed approach comes from two reasons. On the one hand, the Dual-DP-TBD algorithm can realize the detection of the early-exiting target and masked shadow. For example, the early-exiting target in Frame 6 exits the observation scene in Frame 8, while the second detection window integrates from Frame 5 to Frame 9. Therefore, both the classical SW-DP-TBD and R-MF-TBD cannot detect this target while it can be detected using a time-varying detection threshold in (25). There is a masked shadow in Frame 6 whose trajectory passes through a shadow area of a stationary building. R-MF-TBD and Dual-DP-TBD can detect it due to recursive tracking of a known target. On the other hand, fine initial and transferred states obtained by dual-domain supervised resampling improve the tracking performance. The good control ability of false alarm of the proposed approach also comes from two reasons: accurate tracking and elimination of the falsely matched target. A stationary target shadow rotating with the changes of the line of sight may be initialized in the preliminary estimation, but it can be deleted in the fine estimation since there is no Doppler energy in the RD spectrum to match the wrong velocity, as shown in Figs. 11(d) and 13(d). The missing alarm of the proposed approach comes threefold: tracking delay, failed tracking, and failed initialization, which can be seen in Fig. 13(b) and (c). A typical example of tracking delay is given in Frame 12. A target enters the observation scene in Frame 11, however, it will not be detected until Frame 13 (the start frame of the fourth detection window). The false alarm of the proposed approach also comes threefold: error initialization of stationary target shadow, failed tracking, and successive tracking of a false alarm declared in the last window. The above reasons also lead to false alarms and missing alarms in classical SW-DP-TBD and R-MF-TBD algorithms and even become more serious due to inaccurate states and the lack of effective velocity supervision.

Table II illustrates the comparison of detection performances of different methods. The following metrics are used to evaluate the shadow detection performance: 1) the number of missing alarms N_{ma}; 2) the number of false alarms N_{fa}; 3) detection probability P_d that is the ratio of the number of correctly detected shadows to the total number of shadows; 4) average number of false alarms per frame P_{fa} that is the ratio of N_{fa} to the total number of frames; and 5) computational load of DP process. There is a total of 31 trajectories in this observation period which contain 543 moving target shadows. The real frame indices of each trajectory and corresponding detected frame indices of different methods are illustrated in Table III. It can be seen that there are fewer missing alarms and false alarms in the proposed Dual-DP-TBD algorithm compared to other classical DP-TBD algorithms. The background removal and predetection cost 10.6155 s which is the same

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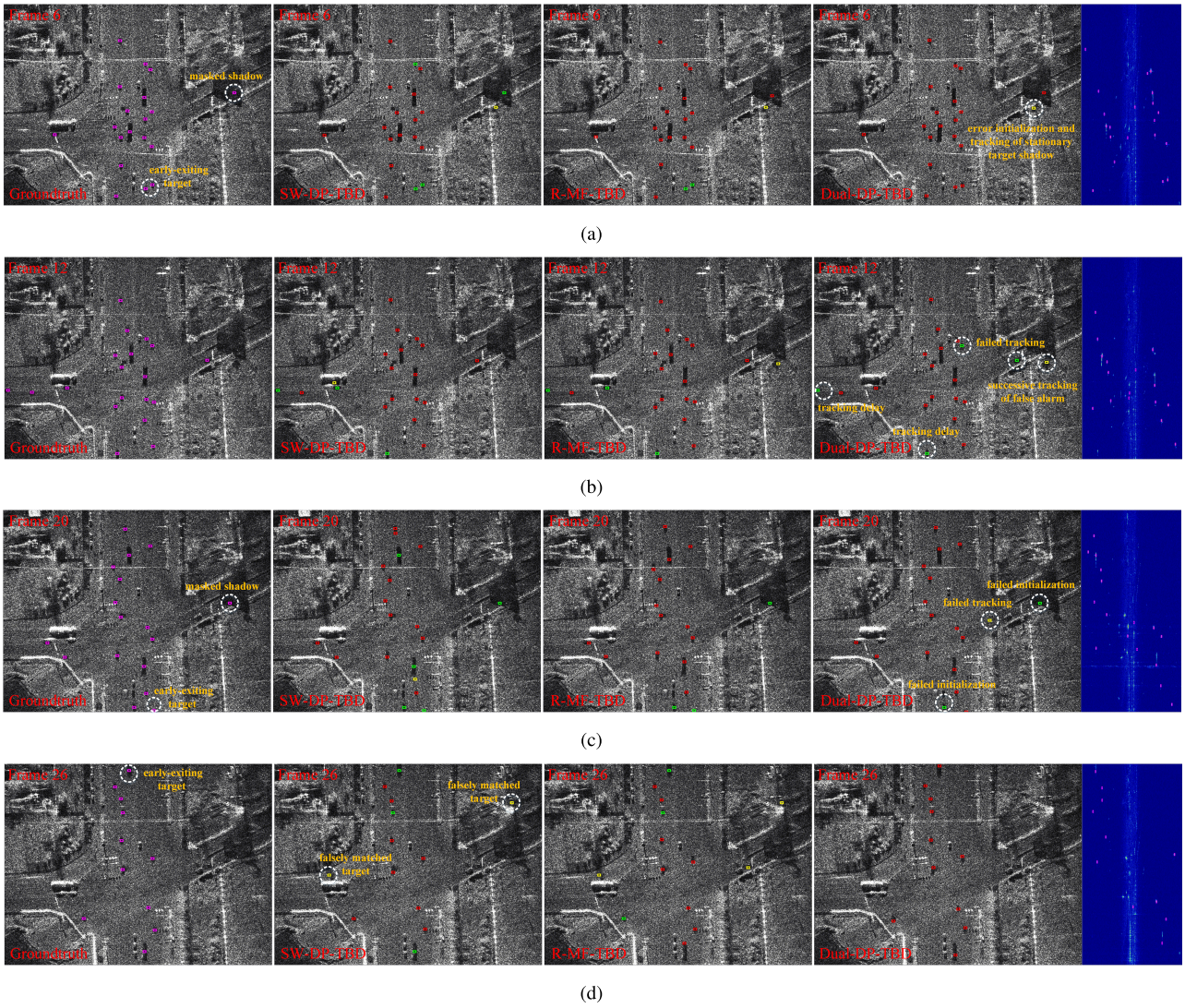


Fig. 13. Comparison of the detection results. (a)–(d) Results of Frames 6, 12, 20, and 26, respectively. In each row, the ground truth and the results of SW-DP-TBD, R-MF-TBD, and the proposed Dual-DP-TBD algorithms are shown from left to right. The correct detection, false alarm, and missing alarm are indicated by red, yellow, and green rectangles, respectively. The results are marked in the original SAR images and suppressed spectra by ADPCA.

TABLE II

COMPARISON OF DETECTION PERFORMANCE OF DIFFERENT METHODS ON MEASURED VIDEO SAR DATA

Approach	Shadows	Missing Alarm N_{ma}	Detection Probability P_d	False Alarm N_{fa}	Average False Alarm P_{fa}	Time Cost
SW-DP-TBD	543	91	0.8324	42	1.2727	58.7680 s
R-MF-TBD		64	0.8821	42	1.2727	62.4380 s
Dual-DP-TBD		**40**	**0.9263**	**26**	**0.7879**	**32.2105 s**

for all DP-TBD algorithms. Table II shows the comparison of computational load of subsequent DP process. The proposed approach has the lowest computational load due to small N_s' although resampling introduces additional computation.

The detection results of typical targets by the proposed Dual-DP-TBD algorithm are presented in Fig. 14, where T6 and T17 move in the range direction while T1, T22, and T23 move in the azimuth direction. In particular, T1 and T23 stop during the illumination, and T22 turns right. Since the Doppler energies of T6 and T17 are outside the clutter band, they can be accurately tracked by dual-domain supervision. Although the Doppler energies of targets that move in the azimuth direction are usually inside the clutter band, the stationary clutter is effectively suppressed by ADPCA. Hence, the Doppler supervision is still effective, provided that the target has a high SCNR. T1 is correctly tracked until it stops; however, it is more complex for T22 and T23. T22 turns right after Frame 12 and its Doppler energy crosses the clutter band in the subsequent frames. In Fig. 14(b), its Doppler energy is smeared and the SCNR is low. In this case, the fine transferred state of T22 cannot cover the correct Doppler energy. Since its radial velocity is small, the slight estimation error has little influence on the detection in the current frame. The proposed Dual-DP-TBD still correctly detects T22 in Frame 13. However, this influence becomes strong with the increase in the number of tracking frames due to velocity

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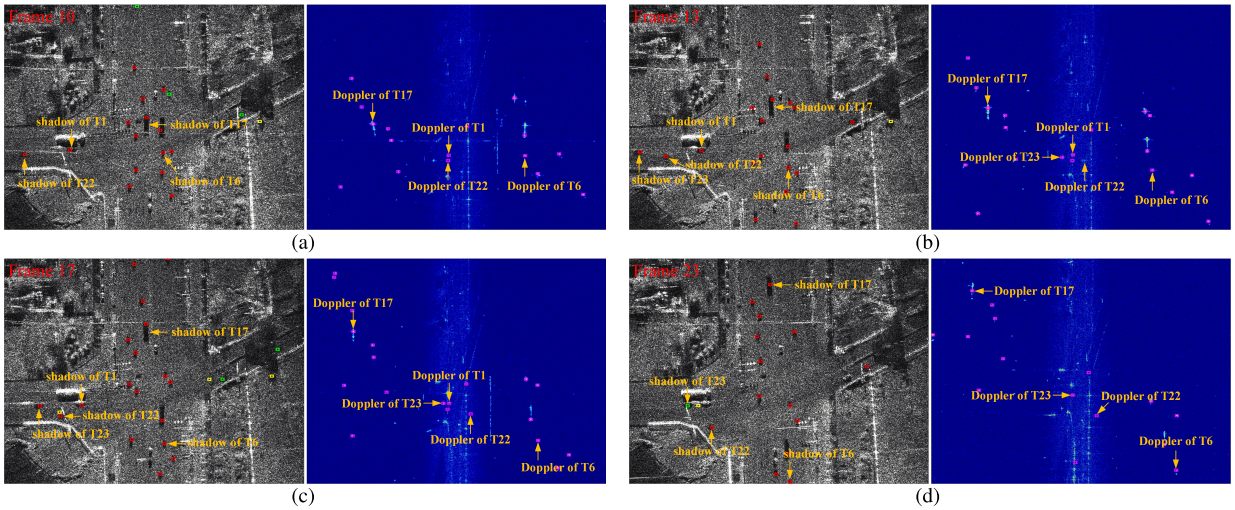


Fig. 14. Detection results of typical targets of the proposed Dual-DP-TBD algorithm. (a)–(d) Results of Frames 10, 13, 17, and 23, respectively.

TABLE III
COMPARISON OF REAL FRAME INDICES AND DETECTED FRAME INDICES

No.	Groundtruth	SW-DP-TBD	R-MF-TBD	Dual-DP-TBD
1	[1,17]	[1,9]∪[13,17]	[1,9]∪[13,17]	[1,17]
2	[1,33]	[1,5]∪[9,21]∪[25,33]	[1,33]	[1,33]
3	[1,31]	[1,3]∪[5,21]∪[25,29]	[1,3]∪[5,29]	[1,8]∪[13,31]
4	[1,29]	[1,14]∪[17,19]∪[21,25]	[1,29]	[1,29]
5	[1,20]	[1,13]	[1,17]	[1,20]
6	[1,23]	[1,21]	[1,21]	[1,23]
7	[1,18]	[1,17]	[1,17]	[1,18]
8	[1,13]	[1,13]	[1,13]	[1,13]
9	[1,8]	[1,5]	[1,5]	[1,8]
10	[1,8]	[1,5]	[1,5]	[1,8]
11	[1,5]	[1,5]	[1,5]	[1,5]
12	[1,4]	undetected	undetected	[1,4]
13	[1,10]	[1,9]	[1,9]	[1,9]
14	[1,17]	[1,17]	[1,17]	[1,17]
15	[1,22]	[1,21]	[1,21]	[1,22]
16	[1,29]	[1,29]	[1,29]	[1,29]
17	[1,27]	[1,18]∪[21,25]	[1,25]	[1,26]
18	[1,31]	[1,30]	[1,29]	[1,31]
19	[1,33]	[1,33]	[1,33]	[1,33]
20	[1,4]	[1,2]	[1,2]	undetected
21	[1,19]	[1,5]∪[9,17]	[1,17]	[1,9]∪[13,15]
22	[8,33]	[9,21]∪[25,33]	[9,25]∪[29,33]	[9,33]
23	[12,25]	[13,25]	[13,25]	[13,22]
24	[30,33]	undetected	undetected	undetected
25	[15,25]	[21,25]	[21,25]	[21,24]
26	[17,33]	[17,21]∪[25,33]	[17,33]	[17,33]
27	[29,33]	[29,33]	[29,33]	[29,33]
28	[5,31]	[5,21]	[5,21]	[5,31]
29	[11,33]	[13,33]	[13,33]	[13,33]
30	[18,25]	[21,25]	[21,25]	[21,25]
31	[27,33]	[29,33]	[29,33]	[29,33]

integration, resulting in a false alarm in Frame 17. Meanwhile, T22 is considered a newborn target in the fifth window and tracked correctly in subsequent frames. The proposed Doppler supervision mainly constrains the radial velocity component. Although the fine transferred state correctly covers the Doppler energy of T23 in Frame 23, the estimated azimuth velocity is not correct, resulting in wrong shadow detection. Nevertheless, the detection probabilities of T1, T22, and T23 are 100%, 96.15%, and 71.43%, respectively. Five false alarms are produced during the tracking of these three targets.

It can be concluded that the proposed Dual-DP-TBD algorithm has shown good performance in moving target detection of video SAR in terms of missing alarm, false alarm, and computational load.

V. CONCLUSION

A Dual-DP-TBD algorithm by embedding the Doppler supervision into an improved DP-TBD algorithm has been presented and used to process the measured data of a UAV-borne multichannel video SAR. An ADPCA is used to suppress the stationary clutter and noise and to improve the SCNR. A fine state estimation method is developed to provide effective velocity guidance by using the Doppler supervision. Fine states can be searched in dual-domain through resampling from random expansion in state initialization and transition. The designed dual-domain merit function is used to simultaneously integrate inverse shadow amplitude in the SAR image and Doppler energy in the RD spectrum. Additionally, it has been verified that the proposed approach can successfully track a time-varying number of targets in continuous measurements.

Compared to other DP-TBD algorithms, this approach exhibits better detection performance and lower computational load when applied to the real data of video SAR.

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